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0 ≤ r < B

Combining these requirements, we will specify the desired input-output relation as

— outputs: A = q X B + r and 0 < r < B.

The comment that starts with — inputs: actually is the property of the given inputs. The comment that starts with — outputs: is the desired relation between the inputs and the outputs. The specification of the algorithm is

1. **divide (A , B)**
2. **-- inputs: A is an integer and B ≠ 0**
3. **-- outputs : A = q X B + r and 0 ≤ r < B**

Specification format: We can write the specification in a standard three part format:

- The name of the algorithm and the inputs.
- Input: the property of the inputs.
- Output: the desired input-output relation.

The first part is the name of the algorithm and the inputs. The second part is the property of the inputs. It is written as a comment which starts with — inputs: The third part is the desired input-output relation. It is written as a comment which starts with — outputs:. The input and output can be written using English and mathematical notation.

Example 6.7. Write the specification of an algorithm for computing the square root of a number.

1. Let us name the algorithm square_root.
2. It takes the number as the input. Let us name the input n. n should not be negative.
3. It produces the square root of n as the output. Let us name the output y. Then n should be the square of y.

Now the specification of the algorithm is

square_root(n)

-- inputs: n is a real number, n ≥ 0.

-- outputs: y is a real number such that $y^2 = n$.

6.5.1 Specification as contract

Specification of an algorithm serves as a contract between the designer of the algorithm and the users of the algorithm, because it defines the rights and responsibilities of the designer and the user.

Ensuring that the inputs satisfy the required properties is the responsibility of the user, but the right of the designer. The desired input-output relation is the responsibility of the designer and the right of the user. Importantly, if the user fails to satisfy the properties of the inputs, the designer is free from his obligation to satisfy the desired input-output relation.